Fiscal Year:	FY 2014	**Task Last Updated:**	FY 06/15/2015
PI Name:	Levine, Benjamin D M.D.		
Project Title:	Cardiac Atrophy and Diastolic Dysfunction During and After Long Duration Spaceflight: Functional Consequences for Orthostatic Intolerance, Exercise Capacity, and Risk of Cardiac Arrhythmias		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	benjaminlevine@texashealth.org	**Fax:**	FY 214 345-4618
PI Organization Type:	UNIVERSITY	**Phone:**	214-345-4619
Organization Name:	The University of Texas Southwestern Medical Center at Dallas		
PI Address 1:	Institute for Exercise and Environmental Medicine (IEEM)		
PI Address 2:	7232 Greenville Ave, Suite 435		
PI Web Page:			
City:	Dallas	**State:**	TX
Zip Code:	75231-5129	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	99-HEDS-03
Start Date:	01/10/2004	**End Date:**	07/31/2014
No. of Post Docs:	3	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Villarreal, Jennifer	**Contact Phone:**	281-483-7306
Contact Email:	jennifer.v311larreal@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS Increment 19 (info from PI 2/2009) NOTE: Project end date is 7/31/2014 per J. Villarreal/JSC (Ed., 10/8/14) NOTE: to be extended per D. Risin/JSC; projected new end date per CoPI M. Bungo is 9/30/2016 (Ed., 4/21/14) NOTE: End date now 3/31/2014 per JSC HRP Master Task List dtd 5/20/2011 (Ed., 7/22/2011) NOTE: End date corrected to 9/30/2012 per J. Dardano/JSC and PI; previous end date showed 1/10/2009 in error (1/2009 information)		
Key Personnel Changes/Previous PI:	PI: Benjamin D. Levine, M.D. ; Co-PI: Michael W. Bungo, M.D. no key personnel changes occurred.		
COI Name (Institution):	Bungo, Michael (CoPI: University of Texas, Houston)		
Grant/Contract No.:	NNJ04HH01A		
Performance Goal No.:			
Performance Goal Text:			
	Cardiac atrophy appears to develop during spaceflight or its ground based analogues, leading to diastolic dysfunction and orthostatic hypotension. Such atrophy also may be a potential mechanism for the cardiac arrhythmias recently identified in some crew members after long duration exposure to microgravity aboard the Mir space station. Recent work by the investigators of this flight experiment has suggested that cardiac atrophy may be progressive, without a clear plateau over at least 12 weeks of bed rest, and thus may be a significant limiting factor for extended duration space missions. The purpose of this experiment is to quantify the extent and time course of cardiac atrophy and identify its mechanisms. The functional consequences of this atrophy also will be determined for cardiac filling dynamics, orthostatic tolerance, exercise tolerance, and arrhythmia susceptibility both in space on the International Space Station (ISS), and following return to Earth. Hypotheses: Hypothesis 1a: Cardiac atrophy occurs in humans after prolonged spaceflight as a function of both the basal hemodynamics of microgravity (reduced wall stress and volume load) and a reduction in physical activity. Hypothesis 1b: Cardiac atrophy develops progressively during long duration spaceflight, but eventually plateaus in response to a new		

Task Description:	steady-state of cardiac work. Hypothesis 2: Cardiac atrophy during prolonged spaceflight reduces ventricular distensibility manifested as decreased end-diastolic volume for any given filling pressure. We propose that this remodeling of the heart limits the utilization of diastolic suction and impairs ventricular relaxation and filling both at rest and during orthostatic stress. This impairment of both static and dynamic diastolic function, in combination with hypovolemia is the primary mechanism leading to a reduced stroke volume during orthostasis after spaceflight. Hypothesis 3: Prolonged spaceflight leads to alterations in ventricular conduction and refractoriness, and an increase in the frequency of arrhythmias. Objectives: Objective 1a: To determine the magnitude of left and right ventricular atrophy associated with long duration spaceflight, and to relate this atrophy to measures of physical activity and cardiac work in-flight. Magnetic resonance imaging (MRI) will be performed pre-and post-flight as the most accurate means of measuring cardiac mass, including sufficient repetitions to document recovery. Objective 1b: To determine the time course and pattern of progression of cardiac atrophy in-flight using cardiac ultrasound. Objective 2: To determine the functional importance of cardiac atrophy for orthostatic tolerance and the regulation of stroke volume, by using a novel combination of innovative, non-invasive imaging techniques to measure the dynamic component of diastole including: a) tissue Doppler imaging to quantify ventricular wall motion with high temporal resolution; b) color Doppler M-mode echocardiography to quantify intraventricular pressure gradients and assess the magnitude of diastolic suction; and c) magnetic resonance imaging with myocardial tagging to quantify the rate of untwisting. Objective 3: To identify changes in ventricular conduction and repolarization during and after long duration spaceflight, and relate these to changes in cardiac mass. Although the direct assessment of arrhythmia risk in otherwise healthy subjects is imprecise, we will obtain hypothesis generating, pilot data using the best non-invasive tests available by measuring: 1) delayed conduction via the signal averaged ECG; 2) dispersion of refractoriness from QT dispersion; and 3) microvolt T wave alternans during routine exercise. New Information Expected: After completion of this study, the clinical expression of cardiac atrophy during long duration spaceflight will be defined clearly, and its significance for orthostatic tolerance and cardiac performance, both systolic and diastolic, will be elucidated. In addition preliminary information will be obtained regarding ventricular conduction and repolarization that may provide either clinical reassurance, or pathophysiolgic insight into the risk for cardiac arrhythmias. Relevance to Space and/or Earth-Based Research: Once the magnitude, time course, and inciting factors for cardiac atrophy have been determined, effective countermeasures currently being developed by the investigators in parallel ground based experiments, may be applied, focused on normalizing cardiac work and volume during long-duration spaceflight. The investigators have recently demonstrated that preventing and/or reversing the atrophy and hypovolemia during short duration head-down tilt bed rest completely eliminates orthostatic intolerance after this ground based analogue. If a similar pathophysiology can be demonstrated after long duration spaceflight, a strong rationale can be made for implementing the same countermeasure during missions aboard the ISS, potentially eliminating a problem, namely post-flight orthostatic hypotension, that has plagued astronauts since the inception of the manned space program. The information obtained from these spaceflight experiments also may be relevant for patients after prolonged confinement to bed rest, or chronic reduction in physical activity, as well as for patients with disease processes that alter cardiac stiffness such as congestive heart failure, ischemic heart disease, and normal aging. See also: https://
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Once the magnitude, time course, and inciting factors for cardiac atrophy have been determined, effective countermeasures currently being developed by the investigators in parallel ground based experiments may be applied, focused on normalizing cardiac work and volume during long-duration spaceflight. The investigators have recently demonstrated that preventing and/or reversing the atrophy and hypovolemia during short duration head-down tilt bed rest completely eliminates orthostatic intolerance after this ground based analogue. If a similar pathophysiology can be demonstrated after long duration spaceflight, a strong rationale can be made for implementing the same countermeasure during missions aboard the ISS, potentially eliminating a problem, namely post-flight orthostatic hypotension, that has plagued astronauts since the inception of the manned space program. The information obtained from these spaceflight experiments also may be relevant for patients after prolonged confinement to bed rest, or chronic reduction in physical activity, as well as for patients with disease processes that alter cardiac stiffness such as congestive heart failure, ischemic heart disease, and normal aging. In practice, this information, in conjunction with work done by the PI funded by NSBRI, has led to a novel treatment for the Postural Orthostatic Tachycardia Syndrome (POTS), and assisted with the rescue of the Chilean Miners.
Task Progress:	Data collection and analyses has been completed. Discussions of the most recent data have been presented at the NASA Investigators' Working Group held in Galveston, Texas during February 2014. A final technical report has been submitted to NASA. Further public data is not available at this time pending publication in peer review journals. Bibliography accompanying this submission describes partial results. The final crewmember of our allotted cohort returned to Earth during May 2013 and completed postflight data collection by the end of June 2013. All ground-based data collections occurred nominally. Inflight data collection was hampered by minor malfunctions of the Cardiopress equipment but no significant data loss resulted.
Bibliography Type:	Description: (Last Updated: 03/19/2026)
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Awards	Bungo MW, Levine BD, PIs: award accepted & presentation by Co-I Jeff Hastings at Plenary Session. "Award accepted for 'Most Compelling Results from the ISS in 2013 – Integrated Cardiovascular Results' Plenary Session. Accepted and presentation by Co-I Jeff Hastings at Plenary Session, 3rd Annual ISS Research & Development Conference of the American Astronautical Society, Chicago, June 2014." Jun-2014